How Large Language Models Answer Questions About Economic Elasticities

A repeated-elicitation study of prompt-conditioned response distributions

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This paper studies the prompt-conditioned response distributions that frontier large language models produce when directly asked about economic elasticities. Rather than asking whether the models can forecast realized outcomes or simulate policy behavior, I ask a narrower question: under a fixed elicitation protocol, what point estimates and uncertainty distributions do the models return? The main dataset contains 6,630 successful runs across 17 frontier models (11 elicited in April 2026 and 6 in July 2026), 26 U.S.-scoped quantities (including a capital-gains convention sibling), and 15 runs per model-quantity cell, all under a single direction-first v4 prompt that embeds neutral sign-convention clarifiers for the two quantities with documented ambiguity. The headline analysis focuses on a nine-elasticity subset of the 13-quantity canonical panel, grouped into labor-and-tax and macro-and-trade subpanels; the panel's four calibration-style parameters enter the stability and robustness checks, and simulation-facing PolicyEngine response coefficients and the capital-gains convention audit appear in separate appendix tables.

Three descriptive findings stand out. First, the model ordering is domain-specific rather than global. On the labor-and-tax subpanel, the most elastic models by average within-quantity absolute-value rank are Claude Sonnet 4.6 and Grok 4.20; on the macro-and-trade subpanel, that ordering changes sharply, with GPT-5.4 mini and Grok 4.3 moving to the top. Second, most labor-and-tax pooled centers fall inside rough review-based benchmark ranges, though the capital-gains realizations elasticity remains the most cross-model-dispersed object within the labor-and-tax subpanel. Third, the capital-gains convention audit — eliciting both parameterizations of the same economic object in parallel under the v4 clarifier — confirms the sign-convention story largely holds up: 16 of 17 models return a negative w.r.t.-tax-rate elasticity paired with a positive w.r.t.-net-of-tax-rate elasticity, consistent with the identity $\varepsilon_\tau = -\tau/(1-\tau) \cdot \varepsilon_{1-\tau}$. The paper's contribution is methodological and descriptive: it offers a reproducible protocol for eliciting economic parameter distributions from LLMs and documents how those prompt-conditioned distributions vary across models and domains.

1 Introduction

Economists use elasticities constantly. They enter sufficient-statistics formulas, optimal tax calculations, quantitative macro calibrations, and applied policy debates. Yet many of the elasticities that matter most are uncertain, interpretation-sensitive, and contested. If large language models are going to be used as policy assistants, research aids, or informal synthetic experts, it matters what they say when asked for those parameters.

The key object in this paper is not forecasting accuracy and not simulated policy behavior. It is a prompt-conditioned elicited response distribution. I ask a model for a quantity such as the Frisch elasticity of labor supply, the elasticity of taxable income, or the Armington elasticity. I require a fixed interpretation, a point estimate, and a distributional summary. Repeating that elicitation many times lets me measure three distinct objects:

- the central estimate the model tends to report
- the uncertainty the model states within a run
- the variation the model exhibits across repeated runs

This is conceptually close to expert elicitation, but with machine respondents rather than human experts. It is also close to the recent literature on LLM uncertainty elicitation, but the target is different. I am not asking for confidence about a ground-truth quiz answer. I am asking what distribution a model returns under a fixed protocol when the target is a contested economic quantity. In that setting, the disagreement itself is often part of the result.

The initial empirical question is narrow: what distributions do leading models return for a common panel of economic elasticities? The elasticity panel is useful because it includes canonical parameters with clear policy relevance and well-known disagreements in the literature. For the labor-supply and tax subset, the policy reading is straightforward: holding welfare weights fixed, lower behavioral elasticities generally imply more room for redistribution, while higher elasticities generally imply larger efficiency costs of redistribution. I do not claim that same monotone policy mapping for every quantity in the panel, and the paper now treats labor-and-tax results separately from macro-and-trade results for exactly that reason.

This paper therefore contributes on three margins. First, it provides a reproducible design for eliciting probabilistic response distributions from LLMs over economic quantities. Second, it documents systematic cross-model differences in central estimates and pooled predictive uncertainty. Third, it shows that these differences are domain-specific: the ordering of models on labor-and-tax elasticities is not the same as the ordering on macro-and-trade elasticities.

2 Related Literature

This project sits at the intersection of four literatures.

First, it is closest in spirit to expert elicitation over uncertain parameters. In economics, the most directly relevant precedent is the ETI elicitation exercise in Designing, Not Checking, for Policy Robustness (2020), which constructs a subjective distribution over the long-run elasticity of taxable income by combining the empirical literature with a survey of economists. More generally, the decision-analysis literature on quantile-based elicitation and "Quantile-Parameterized Distributions for Expert Knowledge Elicitation" (2024) provides a natural methodological precedent for asking directly for quantiles and then fitting a distribution that preserves them.

Second, it relates to the recent literature on uncertainty elicitation from black-box LLMs. Can LLMs Express Their Uncertainty? An Empirical Evaluation of Confidence Elicitation in LLMs (2023) is the clearest baseline for this paper's elicitation design because it explicitly decomposes uncertainty elicitation into prompting, sampling, and aggregation choices. Generating with Confidence (2023) is similarly important because it treats repeated generations as an uncertainty object in their own right rather than only as alternative answers.

Third, the paper is adjacent to the growing literature on calibration and overconfidence in LLM uncertainty reports. LLMs Are Overconfident (2025) and Bayesian Elicitation with LLMs (2026) are especially close to the current design because they ask models for numerical intervals and then evaluate coverage and recalibration. Those papers target out-of-sample predictive reliability against realized truths. Our main estimand is different: the distribution a model manifests over contested economic parameters, whether or not those parameters admit a single clean benchmark truth.

Fourth, the paper borrows from the distribution-aggregation logic used in forecasting platforms. The closest practical analogy is Metaculus, whose FAQ states that for numeric and date questions the community prediction is a weighted average of individual forecaster distributions (Metaculus 2026). That is not the same object as "model belief," but it is a useful precedent for our decision to aggregate run-level distributions rather than average quantiles or endpoints directly.

Finally, the economics side of the project is anchored in review and handbook-style sources rather than in any one calibration tradition. For labor supply, McClelland and Mok (2012) is the key benchmark source because it synthesizes both income and wage elasticities across groups and margins. Similar review-style anchors are used throughout the registry wherever possible.

3 Design

3.1 Prompting target

The main prompt is a memory-only elicitation prompt. It tells the model to answer from background knowledge only, not to use tools or external resources, and not to reconstruct a consensus estimate by conducting a literature review. The prompt fixes the target interpretation of the quantity and requests JSON with:

- interpretation
- point_estimate
- quantiles.p05
- quantiles.p25
- quantiles.p50
- quantiles.p75
- quantiles.p95
- citations
- reasoning_summary

Prompt versions are recorded in the run artifacts and are not pooled across materially different prompt families. The current results therefore combine a main memory-only panel with separately labeled robustness reruns when a quantity needed a targeted clarification.

Two quantities in the registry have well-documented convention ambiguity — the prime-age income elasticity (whether an increase in non-labor income raises or reduces hours is a sign, not a magnitude, question) and the capital-gains realizations elasticity (whether the elasticity is taken w.r.t. the tax rate or the net-of-tax rate flips the sign). For both of these, the v4 main prompt embeds a symmetric direction-first sign clarifier of the form "$\varepsilon > 0$ if and only if … / $\varepsilon < 0$ if and only if …" that enumerates both directions without calling either typical or correct. The clarifier lists the $\varepsilon > 0$ clause first in all cases, which is itself a minor ordering-anchoring choice; a full treatment would randomize clause order across runs. I do not claim the clarifier is fully neutral — any ordering of the clauses can still carry residual anchoring — but it replaces a literature prior with an if-and-only-if definition of what the reported sign means. Other object-definition clarifications — the Armington top-nest clarification and the IES nondurable-consumption clarification — are kept out of the main prompt and reported separately as appendix robustness probes because those clarify object definition rather than sign.

An archived GPT-only robustness arm also allowed web search and code interpreter access on an earlier eight-quantity subset. In realized behavior, tool uptake in that arm was negligible: only 9 / 360 requests used web search, and 0 / 360 used code interpreter. I therefore treat the main design as memory-based elicitation and report the tool-access arm only in the appendix.

3.2 Quantities

The current manuscript uses a 26-quantity U.S.-scoped panel: the 9 canonical elasticities, a capital-gains convention sibling, 3 preference/macro objects, 1 TFP-persistence coefficient, and 12 simulation-facing PolicyEngine response coefficients. Headline model-comparison tables focus on the canonical elasticity subpanel rather than the full mixed panel.

The nine canonical quantities are:

- intertemporal elasticity of substitution
- extensive-margin labor supply elasticity for single mothers
- Frisch elasticity of labor supply for prime-age workers
- income elasticity of labor supply for prime-age workers
- Marshallian wage elasticity of labor supply for prime-age workers
- elasticity of substitution between capital and labor
- capital gains realizations elasticity
- elasticity of taxable income for top earners
- Armington elasticity between imported and domestic goods

The panel also includes 12 simulation-facing labor-supply response coefficients used in PolicyEngine-style microsimulation, including a global substitution elasticity override, 10 primary-earner substitution elasticities by decile, and a secondary-earner substitution elasticity. These are implementation-facing response parameters rather than textbook elasticity objects, so the paper reports them in separate appendix-style tables rather than mixing them into the headline cross-model rankings.

One additional row in the current full panel, labor_supply.policy_response.income_elasticity, is now treated as a labeled legacy quantity rather than an active registry object. It remains in the full comparison data because it was part of the earlier full-panel run, but it is not the preferred canonical income-elasticity measure going forward and it is excluded from the headline tables.

The registry also carries a capital-gains convention sibling: tax.capital_gains_realizations.elasticity.net_of_tax_rate elicits the same economic object as the canonical capital-gains realizations elasticity, but defined w.r.t. the net-of-tax rate $(1-\tau)$ rather than w.r.t. the tax rate τ. The two conventions are related by $\varepsilon_\tau = -\frac{\tau}{1-\tau}\,\varepsilon_{1-\tau}$, so a model that answers consistently across both conventions should report a negative ε_τ paired with a positive $\varepsilon_{1-\tau}$. The sibling is not counted in the nine-elasticity headline subpanels; it appears only in the dedicated convention-audit appendix table. The canonical capital-gains cell in the headline panel continues to use the w.r.t.-tax-rate convention so it stays consistent with the downstream PolicyEngine-US consumer parameter.

3.3 Repeated elicitation and pooling

Each model-quantity cell is elicited 15 times. For a given run r, the prompt elicits five quantiles $q_{r,05}, q_{r,25}, q_{r,50}, q_{r,75}, q_{r,95}$. We convert those quantiles into a run-level distribution G_r using a piecewise-uniform approximation with probability masses $(0.05, 0.20, 0.25, 0.25, 0.20, 0.05)$ on the bins:

- $[L_r, q_{r,05}]$
- $[q_{r,05}, q_{r,25}]$

- $[q_{r,25}, q_{r,50}]$
- $[q_{r,50}, q_{r,75}]$
- $[q_{r,75}, q_{r,95}]$
- $[q_{r,95}, U_r]$

where L_r and U_r are either quantity support bounds or short extrapolations from the outer quantiles when the support is not bounded. For model m and quantity k, the pooled predictive distribution is the equal-weight mixture

$$F_{mk}(x) = \frac{1}{R}\sum_{r=1}^{R} G_{mkr}(x),$$

with $R = 15$ in the main design. The pooled point estimate reported in the tables is the mean of the run-level point estimates, and the pooled 90 percent interval is $[F_{mk}^{-1}(0.05), F_{mk}^{-1}(0.95)]$.

The paper's default interval object is therefore the pooled predictive interval, not a confidence interval for the mean response. The goal is to measure the predictive spread implied by the model's repeated elicited answers, not how precisely we know the mean of those answers. REML and Bayesian hierarchical summaries are implemented in the codebase as secondary estimators, but the headline tables use the equal-weight pooled predictive mixture throughout.

The choice of $R = 15$ is pragmatic rather than theoretically optimal. Appendix Table A1 reports a simple prefix-stability check on the canonical 13-quantity subpanel. Relative to the full 15-run pooled summary, using only the first 10 runs in a cell changes the pooled center by a median of 0.002 and the pooled 90 percent width by a median of 0.006; using only the first 5 runs changes the pooled center by a median of 0.003 and the pooled width by a median of 0.02. Appendix Table A14 attaches Monte Carlo standard errors to the same summaries by resampling the 15 runs within each cell: median center standard errors sit near 0.005 or below for most models, relative width standard errors near 2 to 5 percent, and each model's average width rank carries a narrow 90 percent resampling interval, so the predictive-uncertainty ordering is not an artifact of run-level noise at $R = 15$. None of this proves convergence, but it bounds the sampling error the headline summaries carry.

3.4 Models

The current main panel includes 17 models. Eleven were elicited in the April 2026 rerun:

- gpt-5.4
- gpt-5.4-mini
- gpt-5.4-nano
- claude-opus-4.7
- claude-sonnet-4.6
- claude-haiku-4.5

- gemini-3.1-pro-preview
- gemini-3-flash-preview
- gemini-3.1-flash-lite-preview
- grok-4.20
- grok-4.1-fast

Six models released after that rerun were elicited in July 2026 under the same v4 prompts, quantities, and repeated-run design:

- gpt-5.5
- claude-fable-5
- claude-opus-4.8
- claude-sonnet-5
- gemini-3.5-flash
- grok-4.3

The design is intentionally symmetric across providers: same quantities, same prompt family, same number of repeated runs. Each provider appears at its frontier tier as of its elicitation date; superseded models remain in the panel, which makes within-provider generation-to-generation shifts directly observable. One caution applies to such comparisons: the harness mechanism is confounded with elicitation wave (Appendix Table A16 discloses the full per-model configuration), most sharply for Claude, whose April models ran through a forced-function-call path and whose July models ran through native structured outputs. Appendix Table A17 bounds this concern empirically by re-eliciting an April model (Claude Opus 4.7) under the July mechanism: across the nine canonical elasticities the pooled center moves by at most 0.03 (median 0.00), so mechanism effects appear negligible relative to the cross-model and cross-generation differences the paper describes.

3.5 Generation protocol and exclusions

The generation protocol is fixed across the main no-tools panel.

- Models run at temperature = 1.0 where the API accepts a sampling parameter. Claude Fable 5, Claude Opus 4.8, and Claude Sonnet 5 reject sampling parameters, so their requests carry none and repeated-draw variation comes from default sampling.
- OpenAI models use Chat Completions with strict JSON-schema output and batched draws up to n = 8 per request for cost efficiency.
- Claude models through Opus 4.7 and all Gemini and xAI models run through LiteLLM with one request per run. Claude (through Opus 4.7) and Grok use forced function-call output; Gemini uses forced JSON-object output.
- The July 2026 Claude additions (Fable 5, Opus 4.8, Sonnet 5) run through the native Anthropic API with strict JSON-schema structured outputs and one request per run.

Each keeps its provider-default reasoning configuration: always-on for Fable 5, adaptive for Sonnet 5, off for Opus 4.8.

- The nominal completion budget is 1200 tokens per run for the April panel. Models whose reasoning tokens count against the completion budget receive more headroom (4000 tokens for gemini-3.5-flash and grok-4.3; 32000 for the native-path Claude models); the budget is a truncation guard, not part of the elicitation content.
- Each run is an independent request with no conversational carryover, no tool access, and no manual repair of malformed outputs.
- A run counts as successful if and only if it returns machine-parseable structured output. Failed runs are left missing and excluded from pooled summaries.
- Appendix Table A16 discloses the full per-model harness configuration — provider path, output mechanism, completion budget, sampling regime, reasoning configuration, and API identifier — in one place. Sixteen of seventeen identifiers are floating aliases rather than dated snapshots (only claude-haiku-4.5 pins a dated snapshot), so re-elicitation at a later date may hit updated model builds; the elicitation dates above scope every result.

4 Data Collected So Far

The underlying no-tools v4 data collection contains:

- 17 models (11 elicited April 2026, 6 elicited July 2026)
- 26 quantities (9 canonical elasticities plus 1 capital-gains convention sibling, 3 preference/macro objects, 1 TFP persistence, and 12 PolicyEngine simulation-facing coefficients)
- 15 runs per model-quantity cell
- 6,630 successful runs at 100% parse rate across all cells

Success rates are uniformly 100% under v4: every planned run for every model-quantity cell parsed cleanly into structured JSON. Under the earlier v3 panel a single model (grok-4.20) had a 12% structured-output failure rate that affected 38 runs; that failure mode did not recur in the v4 rerun. In the July 2026 extension, 58 runs (2.3% of the extension) initially failed on infrastructure errors — empty responses from gpt-5.5 exhausting its completion budget on reasoning, missing forced tool calls from grok-4.3, three claude-sonnet-5 responses exceeding a 32,000-token output cap, one transient schema-compilation error, and one safety-classifier false positive — and were re-elicited as fresh independent draws under the identical prompts. A per-cell manifest (results/failure-manifest.csv) records every replaced slot with its error class and replacement request IDs, and the re-elicitation script now archives failed records before replacing them (this archiving postdates the July round, whose audit trail is the manifest plus the retained request logs; because the July replacement happened in place, a per-slot include-versus-exclude sensitivity is not reconstructible for that round). The failures are infrastructure artifacts rather than content, but two caveats keep the claim honest: truncation-type failures correlate with response length, so replacement is not provably independent of elicited values; and the two gpt-5.5 capital-gains cells failed in full at the original 1,200-token budget and were re-elicited entirely at an 8,000-token budget, a disclosed within-cell protocol change. All other affected cells replaced at most 5 of 15 runs — under identical settings for the claude-sonnet-5 and grok-4.3 slots, and at the same raised 8,000-token budget for the two partially affected gpt-5.5 cells.

Tracked API cost at the model-total level for the April panel, sorted by decreasing spend:

- claude-opus-4.7: $8.345
- claude-sonnet-4.6: $4.277
- grok-4.20: $3.448
- gemini-3.1-pro-preview: $3.315
- gpt-5.4: $1.353
- claude-haiku-4.5: $1.203
- gpt-5.4-mini: $0.377
- gemini-3-flash-preview: $0.370
- gemini-3.1-flash-lite-preview: $0.286
- gpt-5.4-nano: $0.118
- grok-4.1-fast: $0.058

And for the July 2026 additions:

- claude-fable-5: $18.001
- claude-opus-4.8: $5.860
- gpt-5.5: $5.768
- gemini-3.5-flash: $4.484
- claude-sonnet-5: $2.878
- grok-4.3: $0.751

Total tracked v4 cost: $23.15 for the April main-panel rerun (April clarify-probe costs for the four per-quantity-fallback models predate the cost-aggregation fix and are excluded; see results/README.md) plus $37.74 for the July additions across their main panel and clarify probes, for $60.89 of tracked spend covering the 6,630 main-panel runs and the July share of the 510 clarify-probe runs. Four April premium-tier models (claude-sonnet-4.6, claude-opus-4.7, gemini-3.1-pro-preview, grok-4.20) were re-elicited quantity-by-quantity to work around a provider-side SSL hang that appeared only when many sequential structured-output calls were pipelined through a single LiteLLM client; the per-quantity subprocess fallback now preserves request-log aggregation, so cost figures for those four models are end-to-end tracked rather than extrapolated. The July additions ran through the same per-quantity path with the fixed cost aggregation. Cost dispersion across models is large: the most expensive model (claude-fable-5, whose always-on reasoning is billed as output tokens) costs roughly 310x the cheapest (grok-4.1-fast), and per-successful-run cost ranges by more than two orders of magnitude between the cheapest and most expensive cells.

5 Main Descriptive Results

5.1 Rankings depend on domain, not just provider

The main descriptive result is not one global model ranking. It is that the ranking changes once the quantities are grouped into economically coherent subpanels.

Table 1 reports the canonical labor-and-tax subpanel: Frisch, income, and Marshallian labor-supply elasticities; the extensive-margin elasticity for single mothers; the elasticity of taxable income; and the capital-gains realizations elasticity. On this subpanel, the highest-elasticity models by average within-quantity absolute-value rank are Claude Sonnet 4.6, Grok 4.20, and Grok 4.3. The lowest-elasticity models are Gemini 3.5 Flash, GPT-5.5, and Gemini 3.1 Flash-Lite.

That ordering is economically interpretable, but only loosely. On this labor-and-tax subset, lower absolute elasticities imply a more redistribution-permissive reading ceteris paribus, while higher absolute elasticities imply larger behavioral costs of redistribution. Under that narrow reading, Gemini 3.5 Flash, GPT-5.5, and Gemini 3.1 Flash-Lite sit on the low-response side of the panel, while Claude Sonnet 4.6 and the two most recent Grok models sit on the high-response side. I treat that ranking only as directional. The more meaningful policy translation is the explicit top-tax exercise below.

Table 1. Labor-and-tax canonical overview.

Note: Canonical labor-and-tax subpanel only: 6 quantities x 17 models x 15 runs. Average ranks are computed within quantity using the absolute value of the pooled point estimate.

Model	Provider	Avg abs-elasticity rank (1=highest)	Avg predictive-uncertainty rank (1=narrowest)	Mean absolute pooled center	Mean pooled 90% width	Success rate	Cost / successful run
Claude Sonnet 4.6	Anthropic	4.92	10.17	0.428	0.978	100.0%	$0.0112
Grok 4.20	xAI	5.75	13.67	0.362	1.245	100.0%	$0.0092
Grok 4.3	xAI	7.33	10.83	0.37	1.125	100.0%	$0.0018
Claude Haiku 4.5	Anthropic	7.67	7.83	0.369	0.911	100.0%	$0.0031
Gemini 3 Flash	Google	8	6.67	0.382	0.892	100.0%	$0.0009
GPT-5.4	OpenAI	8.25	10.5	0.411	1.243	100.0%	$0.0036

Model	Provider	Avg abs-elasticity rank (1=highest)	Avg predictive-uncertainty rank (1=narrowest)	Mean absolute pooled center	Mean pooled 90% width	Success rate	Cost / successful run
Claude Opus 4.7	Anthropic	8.5	6.67	0.394	0.93	100.0%	$0.0221
GPT-5.4 mini	OpenAI	8.5	14.5	0.406	1.452	100.0%	$0.0010
Claude Fable 5	Anthropic	8.83	4.83	0.376	0.812	100.0%	$0.0403
Claude Opus 4.8	Anthropic	8.83	7.5	0.391	0.902	100.0%	$0.0146
GPT-5.4 nano	OpenAI	9	11.17	0.313	1.086	100.0%	$0.0003
Grok 4.1 Fast	xAI	9.5	15.17	0.327	1.422	100.0%	$0.0002
Gemini 3.1 Pro	Google	9.67	5.5	0.371	0.79	100.0%	$0.0091
Claude Sonnet 5	Anthropic	10	6.17	0.367	0.893	100.0%	$0.0068
Gemini 3.1 Flash-Lite	Google	10.42	7.67	0.367	1.022	100.0%	$0.0007
GPT-5.5	OpenAI	13.17	6.33	0.328	0.917	100.0%	$0.0159
Gemini 3.5 Flash	Google	14.67	7.83	0.218	1.059	100.0%	$0.0126

Table 2 reports the macro-and-trade subpanel: the intertemporal elasticity of substitution, the capital-labor substitution elasticity, and the Armington elasticity. Here the ordering changes sharply. GPT-5.4 mini, Grok 4.3, and Grok 4.20 move to the top of the ranking, while Claude Opus 4.7 and Claude Haiku 4.5 move to the bottom. The key point is not that one family is uniformly "more elastic." It is that the ordering is domain-specific.

The pooled predictive-uncertainty ranking also changes across subpanels. In labor-and-tax, Claude Fable 5, Gemini 3.1 Pro, and Claude Sonnet 5 are the tightest models, whereas Grok 4.1 Fast, GPT-5.4 mini, and Grok 4.20 are the widest. In macro-and-trade, Claude Haiku 4.5 becomes the tightest model, while Grok 4.20, Grok 4.3, and GPT-5.4 mini are the widest. This is pooled predictive uncertainty, not calibrated confidence.

The macro-and-trade subpanel contains only three quantities, so the ranking is easily overturned by a single object. The leave-one-provider-out robustness (Appendix Table A3) shows Spearman ρ stays above 0.98 even on this subpanel, but the ranking should be read as directional rather than fine-grained. The Armington clarification evidence (Appendix Table A7) reinforces that this subpanel's ordering is particularly sensitive to object definition.

Table 2. Macro-and-trade canonical overview.

Note: Canonical macro-and-trade subpanel only: 3 quantities x 17 models x 15 runs. Average ranks are computed within quantity using the absolute value of the pooled point estimate.

Model	Provider	Avg abs-elasticity rank (1=highest)	Avg predictive-uncertainty rank (1=narrowest)	Mean absolute pooled center	Mean pooled 90% width	Success rate	Cost / successful run
GPT-5.4 mini	OpenAI	4.33	14.5	1.241	3.491	100.0%	$0.0009
Grok 4.3	xAI	4.5	15.67	1.509	3.887	100.0%	$0.0018
Grok 4.20	xAI	4.67	16	1.257	3.986	100.0%	$0.0089
GPT-5.4 nano	OpenAI	5.67	10.67	1.256	2.696	100.0%	$0.0003
GPT-5.4	OpenAI	6	6.33	1.267	2.393	100.0%	$0.0033
Gemini 3.5 Flash	Google	6.67	9.33	1.153	2.473	100.0%	$0.0104
Gemini 3.1 Pro	Google	8	9	1.27	2.506	100.0%	$0.0091
Claude Sonnet 4.6	Anthropic	8.83	8.83	1.16	2.724	100.0%	$0.0109
Claude Sonnet 5	Anthropic	9.33	6.33	1.433	2.948	100.0%	$0.0064
GPT-5.5	OpenAI	10	10.67	1.121	2.918	100.0%	$0.0101
Gemini 3.1 Flash-Lite	Google	10.33	7.33	1.164	2.943	100.0%	$0.0007
Claude Opus 4.8	Anthropic	10.83	6.17	1.167	2.737	100.0%	$0.0137
Grok 4.1 Fast	xAI	11.5	8.17	0.907	1.785	100.0%	$0.0001
Claude Fable 5	Anthropic	12	6.33	1.041	2.127	100.0%	$0.0397

Model	Provider	Avg abs-elasticity rank (1=highest)	Avg predictive-uncertainty rank (1=narrowest)	Mean absolute pooled center	Mean pooled 90% width	Success rate	Cost / successful run
Gemini 3 Flash	Google	12.33	8.67	0.94	2.632	100.0%	$0.0009
Claude Haiku 4.5	Anthropic	13.5	2	0.881	1.389	100.0%	$0.0031
Claude Opus 4.7	Anthropic	14.5	7	0.878	2.224	100.0%	$0.0211

5.2 Most labor-and-tax centers lie inside rough review ranges

Table 3 compares the labor-and-tax pooled centers to rough review-based benchmark intervals. These are not benchmark truths; they are hand-coded literature anchors intended to check whether the elicited centers are at least living in the right neighborhood.

For most labor-and-tax quantities, the answer is yes. Every one of the 17 models falls inside the rough benchmark range for the Frisch elasticity. 16 / 17 lie inside for capital gains realizations, the uncompensated wage elasticity, and the extensive-margin single-mother elasticity, and 14 / 17 for the canonical income elasticity. The sign-convention clarifier in v4 eliminated the clear wrong-sign centers that appeared in the earlier v3 panel, though one April model still sits at an economically null income elasticity (gpt-5.4-nano at -0.001). One July addition reintroduces sign instability in a sharper form: gemini-3.5-flash is bimodal on both sign-sensitive quantities rather than centered near zero. On capital gains realizations, 10 of its 15 runs are negative (between -0.7 and -0.2) and 5 are large and positive (between +0.4 and +1.3), so the mean center of +0.010 is sign cancellation across modes (the run-level median is -0.40); on the income elasticity, 11 runs are mildly negative and 4 are positive, again yielding a near-zero mean (+0.011) that no individual run states. Nine of the fifteen capital-gains runs and five of the fifteen income-elasticity runs also required quantile repair, concentrated in exactly these cells. The clarifier therefore reduces but does not eliminate sign instability in new model generations, and a near-zero mean on a sign-sensitive quantity should be read as a symptom of mode-mixing rather than as a stated belief. gpt-5.5 also centers just above the income-elasticity band at -0.035. ETI remains a quantity with upper-edge pressure: 15 / 17 models fall inside the rough range, with the same two April models (claude-haiku-4.5 at 0.507, gpt-5.4-nano at 0.552) just above the review band's upper anchor of 0.5. The band itself sits in the upper half of the Saez, Slemrod, and Giertz (2012) survey range of 0.12 to 0.40, extended to 0.5 to reflect higher top-earner estimates (Gruber and Saez 2002); several models (Claude Sonnet 4.6 at 0.500, Grok 4.20 at 0.500, and now Claude Sonnet 5 at 0.473) concentrate at or near that upper anchor.

That even the benchmark-consistent models lean toward the top of the review range is itself a finding about the models' elicited ETI distributions.

This benchmark table is useful because it distinguishes two kinds of disagreement. On many quantities, the models disagree with each other but still sit inside a conventional literature neighborhood. On a few quantities, the disagreement includes benchmark outliers. Appendix Table A6 sharpens the same point from a different angle: on every canonical quantity, cross-model spread in pooled centers remains smaller than the average pooled 90 percent width. Within the labor-and-tax subpanel the largest spread-to-width ratio occurs for the capital gains realizations elasticity (0.52); across the full canonical panel, TFP persistence (0.65) and the intertemporal elasticity of substitution (0.55) rank higher still.

Table 3. Rough literature comparison for the labor-and-tax subpanel.

Note: Rough review-based benchmark intervals for the canonical labor-and-tax subpanel. These intervals are hand-coded literature anchors rather than benchmark truths.

Quantity	Rough review range	Models in range	Model min center	Model max center	Benchmark sources
Capital gains realizations elasticity	[-1, -0.2]	16 / 17	-0.93	0.01	Dowd, McClelland, and Muthitacharoen 2015; Burman and Randolph 1994; CBO/JCT medium-run convention
Elasticity of taxable income	[0.25, 0.5]	15 / 17	0.35	0.552	Gruber and Saez 2002 (high-income estimates); Saez, Slemrod, and Giertz 2012 (survey range 0.12-0.40, upper half)

Quantity	Rough review range	Models in range	Model min center	Model max center	Benchmark sources
Employment participation elasticity of single mothers	[0.3, 1]	16 / 17	0.213	0.713	Chetty, Guren, Manoli, and Weber 2013 (elasticity implied by Eissa and Liebman 1996); Meyer and Rosenbaum 2001
Frisch elasticity of labor supply	[0.25, 0.75]	17 / 17	0.283	0.593	CBO 2012; Peterman 2016; lifecycle and macro-calibration literature
Income elasticity of labor supply	[-0.15, -0.05]	14 / 17	-0.107	0.011	CBO 2012; Blundell and MaCurdy 1999; Imbens, Rubin, and Sacerdote 2001 (marginal propensity to earn, converted to an elasticity)
Uncompensated wage elasticity of labor supply	[0.05, 0.3]	16 / 17	0.04	0.16	CBO 2012; Blundell and MaCurdy 1999

5.3 A utilitarian sufficient-statistics top-tax exercise

The labor-and-tax subpanel also permits one concrete policy translation. For the elasticity of taxable income, I map each model's pooled ETI distribution into an optimal top marginal tax rate using the standard sufficient-statistics formula in the top bracket (Saez 2001):

$$\tau^* = \frac{1-\bar{g}}{1-\bar{g}+ae},$$

with a Pareto parameter estimated from the weighted top 1 percent tail of tax-unit adjusted gross income in PolicyEngine's certified microdata (PolicyEngine Team 2024).[1] In the current build, that estimate is $a = 1.621$. For the headline column I assume log utility over consumption, $u(c) = \log c$, and weight top-bracket earners by their average marginal utility normalized to the marginal utility of the earner at the top-bracket threshold. Under a Pareto(a) tail and CRRA(γ) utility that threshold-normalized weight is

$$\bar{g} = \frac{a}{a+\gamma} = \frac{1.621}{2.621} \approx 0.618,$$

so the headline mapping becomes

$$\tau^* = \frac{0.382}{0.382 + 1.621e}.$$

Two caveats on the welfare weight. First, it is a normalization choice, not an implication of the elicited data: the standard population-normalized utilitarian benchmark drives the top weight toward zero as top incomes grow far beyond the population mean, collapsing the formula to the revenue-maximizing (Diamond-Saez) rate $\tau^* = 1/(1+ae)$ (Diamond 1998; Saez 2001). Table 4 reports that $\bar{g} \to 0$ benchmark alongside the headline column; at the elicited ETI medians it runs from 53.1% to 63.8%, so the level of the headline rates — though not the cross-model ordering — is reduced by roughly 40 percent by the threshold normalization. Second, calling the headline column "utilitarian" without qualification would overstate it; it is a utilitarian mapping under a specific, disclosed normalization.

This is still intentionally stylized. It uses only the ETI and therefore does not absorb the rest of the labor-supply block, but it provides a common public-finance object that can be compared

[1] The weighted top-1-percent AGI threshold is \$725,533 in the current build (with a tail mean of \$1,894,129). This is the top-1-percent tax-unit cutoff in the microdata, not the statutory top-bracket edge, which is \$640,600 for single filers and \$768,700 for married-filing-jointly under TCJA-extended / OBBBA parameters in 2026. Robustness to threshold choice maps onto robustness to the Pareto parameter a; see Appendix Table A13. The exercise is not tied to any particular statutory bracket — the ETI-based top-rate formula is a common public-finance object that can be compared across models regardless of where the "top" is drawn. When the microdata calibration is unavailable, the build falls back to $a = 1.5$ and prints a warning; every number in this section comes from the microdata build committed with the paper.

across models. The microdata enter only through the top-tail calibration, not through a full reform simulation. For this policy mapping, I truncate ETI below at zero so the standard sufficient-statistics interpretation remains well defined. Every model's pooled ETI p05 exceeds 0.10 in Table 4, so the truncation affects negligible mass in practice.

Appendix Table A12 shows why I do not replace this benchmark with a static flat-tax-plus-demogrant microsimulation. Absent behavioral responses or leisure in the objective, that kind of exercise is better read as a distributional frontier than as a meaningful optimal-tax calculation.

The richer point is not just the spread in medians but the propagated uncertainty. (Table 4 keys off pooled mixture medians, which differ slightly from the mean centers quoted elsewhere in the text — for example, Claude Haiku 4.5's ETI median of 0.502 versus its mean center of 0.507.) Under this threshold-normalized log-utility mapping with a microdata-calibrated Pareto tail, the implied optimal top-rate median ranges from 30.1% for GPT-5.4 nano to 40.2% for Gemini 3.1 Pro, with GPT-5.4 at 35.9%. That is a 10.1 percentage-point cross-model spread in medians. But the within-model uncertainty bands are much wider: top-rate 90 percent intervals are roughly 42 to 52 percentage points wide across models, partly because several models place non-trivial elicited mass on top-earner ETIs above 1, well beyond the empirical literature. So the exercise cuts both ways. It shows that elicited ETI differences do map into meaningfully different central policy conclusions, but it also shows that any single model's implied optimal-tax recommendation remains very noisy under the paper's own uncertainty object.

Appendix Table A13 reports the same median mapping under alternative values of the Pareto tail parameter a and the CRRA curvature γ. Moving a from the baseline 1.621 to 1.3 shifts every model's implied top rate up by 7.9 to 8.6 percentage points, and moving a to 1.7 shifts every model down by 1.6 to 1.9 percentage points — the cross-model ordering is invariant under these shifts. Replacing log utility with CRRA at $\gamma = 2$ (holding a at the baseline) lowers the threshold-normalized welfare weight from $\bar{g} = 0.618$ to $\bar{g} = 0.448$ and raises every model's top rate by 8.3 to 9.1 percentage points, again without re-ordering the models.

Table 4. Optimal top-tax exercise from ETI distributions.

Note: Toy public-finance mapping from each model's pooled ETI distribution to an optimal top marginal tax rate under the Saez top-bracket formula tau* = (1 - g_bar) / (1 - g_bar + a e), with a Pareto parameter a = 1.621 estimated from the weighted top 1% tax-unit AGI tail in PolicyEngine's certified microdata (threshold $725,533, tail mean $1,894,129). The welfare weight g_bar = a / (a + gamma) = 0.618 is the average marginal utility of top-bracket earners under CRRA (gamma = 1) utility and a Pareto(a) income tail, normalized to the marginal utility of the earner at the top-bracket threshold. It is a threshold-normalized weight, not the population-normalized utilitarian weight, which would drive g_bar toward zero; the Revenue-max column reports that g_bar -> 0 (Diamond-Saez revenue-maximizing) benchmark tau* = 1 / (1 + a e) at the ETI median. ETI is truncated below at zero for this policy mapping.

Model	ETI median [90%]	Top rate median [90%]	Revenue-max median	Top-rate 90% width (pp)
Gemini 3.1 Pro	0.351 [0.103, 0.868]	40.2% [21.3%, 69.5%]	63.8%	48.2
Gemini 3.5 Flash	0.357 [0.102, 0.832]	39.8% [22.0%, 69.8%]	63.4%	47.7
GPT-5.5	0.369 [0.122, 0.994]	39.0% [19.1%, 65.9%]	62.6%	46.8
Claude Opus 4.8	0.383 [0.105, 0.993]	38.0% [19.2%, 69.1%]	61.7%	49.9
Gemini 3.1 Flash-Lite	0.389 [0.101, 0.951]	37.7% [19.8%, 69.9%]	61.3%	50.1
Claude Opus 4.7	0.400 [0.123, 0.993]	37.0% [19.2%, 65.7%]	60.7%	46.5
Gemini 3 Flash	0.400 [0.131, 1.040]	37.0% [18.5%, 64.3%]	60.7%	45.8
Grok 4.1 Fast	0.400 [0.109, 1.191]	37.0% [16.5%, 68.4%]	60.7%	51.9
GPT-5.4	0.420 [0.150, 0.996]	35.9% [19.1%, 61.1%]	59.5%	42
GPT-5.4 mini	0.437 [0.116, 1.377]	35.0% [14.6%, 67.1%]	58.6%	52.5
Claude Fable 5	0.437 [0.151, 1.076]	35.0% [17.9%, 60.9%]	58.6%	42.9
Grok 4.3	0.439 [0.133, 1.091]	34.9% [17.8%, 64.0%]	58.4%	46.2
Claude Sonnet 5	0.471 [0.145, 1.146]	33.3% [17.0%, 61.9%]	56.7%	44.9
Claude Sonnet 4.6	0.500 [0.138, 1.282]	32.0% [15.5%, 63.1%]	55.2%	47.6
Grok 4.20	0.500 [0.174, 1.243]	32.0% [15.9%, 57.5%]	55.2%	41.6
Claude Haiku 4.5	0.502 [0.145, 1.299]	31.9% [15.3%, 61.9%]	55.1%	46.6
GPT-5.4 nano	0.546 [0.151, 1.461]	30.1% [13.9%, 60.9%]	53.1%	47

5.4 Convention clarifications in the main prompt

Two quantities with well-documented convention ambiguity have direction-first sign clarifiers embedded in the v4 main prompt: the prime-age income elasticity (whether extra non-labor income raises or reduces hours is a sign question) and the capital-gains realizations elasticity (whether epsilon is taken w.r.t. the tax rate tau or the net-of-tax rate (1 - tau) flips the sign). The clarifier does not name which direction is conventional — it just says "$\varepsilon > 0$ if and only if X raises Y; $\varepsilon < 0$ if and only if X reduces Y" for each quantity, which fixes the sign-of-reported-number $\leftrightarrow$ behavioral-direction mapping without asserting which direction is empirically right. The clarifier lists the $\varepsilon > 0$ clause first in all cases, which is itself a minor ordering-anchoring choice; a full treatment would randomize clause order across runs. Under v4, the canonical income elasticity center is negative for 16 of 17 models — the exception is gemini-3.5-flash's bimodal +0.011 mean discussed above — and inside the review band for 14 of 17. The other two out-of-band centers are negative but economically small: gpt-5.4-nano at -0.001 and gpt-5.5 at -0.035, both above the band's upper edge. The Armington and IES clarifications continue to produce meaningful cross-prompt deltas (Appendix Tables A7, A8). The capital-gains convention audit (Appendix Table A9) elicits both parameterizations of the same economic object to test whether each model answers consistently across conventions.

A second prompt-sensitivity check on the macro-and-trade side confirms the same methodological point: an 11-model follow-up on the Armington elasticity with an explicit top-level-import-versus-domestic clarification shifts pooled centers downward for most models (Appendix Table A7). Some cross-model disagreement reflects object-definition differences the base prompt underspecifies rather than substantive belief disagreement.

6 Interpretation

The paper's main claim is not that one model is correct and not that the protocol recovers latent stable priors. The protocol identifies prompt-conditioned response distributions under a fixed elicitation design. That is narrower than "belief" in a deep psychological sense, but it is still a useful object for users who interact with these systems through prompts.

That matters for three reasons.

First, domain-specific ordering means users should not talk about models as if they had one general "elasticity ideology." A model can look relatively high-response on labor-and-tax quantities and relatively low-response on macro-and-trade quantities, or vice versa.

Second, pooled predictive uncertainty is itself informative. Some models behave as if the relevant literature is tight and settled; others behave as if the same quantities are wide-open or as if their own answers are unstable across repeated runs. Users interacting with these models will experience that difference directly. One caveat applies to the three no-sampling-parameter Claude models (Fable 5, Opus 4.8, Sonnet 5), whose between-run variation arises under provider-default sampling rather than the panel's temperature = 1.0; Appendix Table A15 shows this matters little in practice, because the between-run component contributes a median of 0 to 2 percent of pooled predictive variance for every model — the widths are dominated by the models' own stated quantiles, which the sampling regime does not touch.

Third, prompt sensitivity is part of the estimand rather than a nuisance to hide. The income-elasticity sign clarification shows that a fixed protocol can still be identifying a mixture of economic judgment and convention handling. That is not a reason to abandon the exercise; it is a reason to describe the object precisely and to invest in quantity-specific prompt design.

This also clarifies the relationship between the present paper and the LLM uncertainty literature discussed above. Papers such as Can LLMs Express Their Uncertainty? An Empirical Evaluation of Confidence Elicitation in LLMs (2023), Generating with Confidence (2023), and LLMs Are Overconfident (2025) mostly evaluate whether uncertainty reports line up with eventual correctness. By contrast, this paper asks how models respond when the target is a contested economic parameter and the disagreement itself is economically interesting. In that sense, the project is closer to synthetic expert elicitation than to benchmark evaluation.

7 Limitations

This manuscript should be read as an initial results paper, not a final measurement of latent model beliefs.

The current limitations are straightforward:

- the protocol identifies prompt-conditioned responses, not stable latent priors
- the main panel is memory-only in design and almost entirely tool-free in realized behavior; retrieval-augmented and tool-using arms are still secondary
- the current full dataset still contains simulation-facing response coefficients and one labeled legacy row, even though the headline tables now separate them
- the current paper focuses on elasticities, not the broader OG-USA-style parameter set
- the interval analysis is descriptive; calibration against resolved numeric tasks is still modular but secondary
- citation strings are noisy and need normalization before they can support serious bibliometric analysis

8 Appendix

8.1 Stability check

Appendix Table A1 reports a simple stability check for the choice of R = 15. Across the canonical 13-quantity subpanel (221 cells at every prefix length), the median absolute difference between the first 10 runs in a cell and the full 15-run pooled center is only 0.002; the corresponding median difference in pooled 90 percent width is 0.006. Using only the first 5 runs is noisier, but still modest at the median.

Appendix Table A1. Prefix stability for the canonical subpanel.

Note: Prefix stability on the 13-quantity canonical subpanel. Rows compare pooled summaries from the first 5 or 10 runs in a cell to the full 15-run pooled summary.

Runs used	Cells compared	Median abs change in pooled center	90th pct abs change in pooled center	Median abs change in pooled width	90th pct abs change in pooled width
5	221	0.003	0.04	0.02	0.225
10	221	0.002	0.023	0.006	0.092
15	221	0	0	0	0

8.2 Pooling robustness

Appendix Table A2 compares model-level predictive-uncertainty rankings under three interval constructions: the headline pooled mixture interval, a REML random-effects predictive interval, and a Bayesian hierarchical predictive interval. The rankings move somewhat, but the changes are not large enough to overturn the broad qualitative picture. Claude Haiku 4.5, Claude Opus 4.7, Gemini 3.1 Pro, and Gemini 3 Flash remain among the tighter models across all three constructions, while GPT-5.4 mini, GPT-5.4 nano, and Grok 4.20 remain toward the wide end. The largest rank spread is 1.85 positions for Claude Sonnet 4.6.

Appendix Table A2. Predictive-uncertainty ranking under alternative pooling methods.

Note: Canonical 13-quantity subpanel. Average predictive-uncertainty ranks are computed under the headline pooled mixture interval, the REML predictive interval, and the Bayesian predictive interval.

Model	Avg pooled rank	Avg REML rank	Avg Bayes rank	Max rank spread
Claude Fable 5	5.15	5.15	5.08	0.08
Claude Haiku 4.5	6.38	5	5.08	1.38
Claude Opus 4.7	6.38	8.08	7.62	1.69
Claude Sonnet 5	6.38	7.23	7.15	0.85
Gemini 3.1 Pro	7.27	6.08	6.54	1.19
Claude Opus 4.8	7.35	8.96	8.85	1.62
Gemini 3 Flash	7.85	7.46	7.15	0.69
Gemini 3.5 Flash	8.31	9.46	9.92	1.62
Claude Sonnet 4.6	8.85	10.69	10.23	1.85
Gemini 3.1 Flash-Lite	8.85	8.46	8.46	0.38
GPT-5.5	9.46	10.31	9.92	0.85
GPT-5.4	9.69	10.69	10.46	1

Model	Avg pooled rank	Avg REML rank	Avg Bayes rank	Max rank spread
GPT-5.4 nano	11	9.38	10.77	1.62
Grok 4.1 Fast	11.35	10.15	9.92	1.42
Grok 4.3	11.38	10.65	10.54	0.85
GPT-5.4 mini	13.58	12.23	12.54	1.35
Grok 4.20	13.77	13	12.77	1

8.3 Leave-one-provider-out stability

Appendix Table A3 asks whether the subpanel rankings are being driven by any single provider family. The answer is mostly no. Across both the labor-and-tax and macro-and-trade subpanels, the leave-one-provider-out Spearman correlation with the full-panel ranking stays between 0.982 and 0.999. Some top spots do flip when one provider is removed, especially on the smaller three-quantity macro-and-trade panel, but the broad ordering is fairly stable.

Appendix Table A3. Leave-one-provider-out ranking sensitivity.

Note: Leave-one-provider-out sensitivity of the average absolute-elasticity ranking on the labor-and-tax and macro-and-trade canonical subpanels.

Subpanel	Omitted provider	Spearman rho	Top retained model, full panel	Top retained model, leave-out	Max avg-rank shift
Labor/tax	Anthropic	0.982	Grok 4.20	Grok 4.20	5.5
Labor/tax	Google	0.996	Claude Sonnet 4.6	Claude Sonnet 4.6	2.5
Labor/tax	OpenAI	0.993	Claude Sonnet 4.6	Claude Sonnet 4.6	3.333
Labor/tax	xAI	0.998	Claude Sonnet 4.6	Claude Sonnet 4.6	2.5
Macro/trade	Anthropic	0.997	GPT-5.4 mini	GPT-5.4 mini	2.833
Macro/trade	Google	0.995	GPT-5.4 mini	GPT-5.4 mini	3.667
Macro/trade	OpenAI	0.992	Grok 4.3	Grok 4.20	3.667
Macro/trade	xAI	0.999	GPT-5.4 mini	GPT-5.4 mini	2.667

8.4 Alternative quantile-to-distribution rule

Appendix Table A4 replaces the headline piecewise-uniform reconstruction with a cruder transformed-normal approximation calibrated to p05, p50, and p95 for each run. The resulting predictive-uncertainty ranks move, but not dramatically. The largest average rank change is 1.88 positions for Claude Sonnet 4.6; most models move by less than one rank position. That does not prove the headline reconstruction is uniquely correct, but it suggests the main uncertainty ordering is not an artifact of one specific quantile-bin interpolation rule.

Appendix Table A4. Sensitivity to the within-run distribution rule.

Note: Canonical 13-quantity subpanel. The alternative rule approximates each run as a transformed normal calibrated to p05, p50, and p95 instead of the headline piecewise-uniform quantile-bin reconstruction.

Model	Avg piecewise-uniform rank	Avg transformed-normal rank	Rank shift
Claude Fable 5	5.15	5.69	0.54
Claude Opus 4.7	6.38	8.15	1.77
Claude Haiku 4.5	6.38	5	-1.38
Claude Sonnet 5	6.38	7.08	0.69
Gemini 3.1 Pro	7.27	6.38	-0.88
Claude Opus 4.8	7.35	9.12	1.77
Gemini 3 Flash	7.85	7.31	-0.54
Gemini 3.5 Flash	8.31	8.77	0.46
Gemini 3.1 Flash-Lite	8.85	8.23	-0.62
Claude Sonnet 4.6	8.85	10.73	1.88
GPT-5.5	9.46	10.15	0.69
GPT-5.4	9.69	9.92	0.23
GPT-5.4 nano	11	10.38	-0.62
Grok 4.1 Fast	11.35	9.85	-1.5
Grok 4.3	11.38	10.62	-0.77
GPT-5.4 mini	13.58	12.69	-0.88
Grok 4.20	13.77	12.92	-0.85

8.5 Tool-access robustness

The paper's main estimand is a memory-based elicitation design. To check whether that framing is misleading in practice, I also ran an earlier GPT-only robustness arm in which the models were given access to web search and code interpreter tools. That arm covered an earlier eight-quantity subset and therefore is not directly comparable to the current 26-quantity main panel. Its value is narrower: it shows whether allowing tools materially changed realized behavior.

In that archived tool-access arm, actual tool uptake was rare. Across all 360 requests, web search was used in only 9 cases (2.5%), and code interpreter was used in 0 cases. Full GPT-5.4 never used a tool; GPT-5.4 mini used web search in 2 / 120 requests; and GPT-5.4 nano used web search in 7 / 120 requests. The appendix table below summarizes those counts.

Appendix Table A5. Tool use in the archived GPT-only tool-access arm.

Note: Archived GPT-only robustness arm with full web and code-interpreter access on the earlier 8-quantity panel (8 quantities x 15 runs x 3 GPT models = 360 requests). In realized behavior, tool uptake was rare and code interpreter was never used.

Model	Requests	Requests with web use	Share with web use	Total web calls	Requests with code use	Share with code use	Total code calls
GPT-5.4	120	0	0.0%	0	0	0.0%	0
GPT-5.4 mini	120	2	1.7%	2	0	0.0%	0
GPT-5.4 nano	120	7	5.8%	7	0	0.0%	0
All GPT models	360	9	2.5%	9	0	0.0%	0

8.6 Canonical quantity disagreement

Appendix Table A6 reports quantity-level disagreement on the canonical 13-quantity panel. The final column scales cross-model spread by the average pooled 90 percent width. On this metric, no canonical quantity has cross-model spread larger than average within-model predictive width.

Appendix Table A6. Canonical quantity disagreement.

Note: Canonical elasticity subpanel only, sorted by cross-model spread in pooled point estimates. Simulation-facing PolicyEngine coefficients are broken out in separate appendix-style tables.

Quantity	Lowest model	Lowest center	Highest model	Highest center	Spread	Mean pooled 90% width	Spread / mean width
Armington elasticity	Claude Haiku 4.5	1.5	Claude Sonnet 5	3.18	1.68	5.312	0.316
Intertemporal elasticity of substitution	Claude Fable 5	0.5	Gemini 3.1 Pro	1.467	0.967	1.763	0.548
Capital gains realizations elasticity	GPT-5.4 mini	-0.93	Gemini 3.5 Flash	0.01	0.94	1.823	0.516
Employment participation elasticity of single mothers	GPT-5.4 nano	0.213	GPT-5.4	0.713	0.5	1.223	0.409
Coefficient of relative risk aversion	Claude Haiku 4.5	1.567	Claude Fable 5	2	0.433	7.644	0.057
Frisch elasticity of labor supply	Claude Haiku 4.5	0.283	GPT-5.4 nano	0.593	0.31	1.261	0.246
Elasticity of substitution between capital and labor	Claude Opus 4.7	0.6	GPT-5.4 mini	0.883	0.283	1.017	0.278
Elasticity of taxable income	Gemini 3.1 Pro	0.35	GPT-5.4 nano	0.552	0.202	0.995	0.203

Quantity	Lowest model	Lowest center	Highest model	Highest center	Spread	Mean pooled 90% width	Spread / mean width
TFP persistence	GPT-5.4 nano	0.823	Gemini 3.5 Flash	0.955	0.132	0.202	0.653
Uncompensated wage elasticity of labor supply	Gemini 3.1 Pro	0.04	Grok 4.20	0.16	0.12	0.586	0.205
Income elasticity of labor supply	Grok 4.20	-0.107	Gemini 3.5 Flash	0.011	0.118	0.351	0.335
Capital share in production	Claude Haiku 4.5	0.307	Claude Fable 5	0.343	0.036	0.186	0.194
Annual discount factor	Grok 4.20	0.959	Claude Sonnet 5	0.965	0.006	0.086	0.07

8.7 Armington clarification follow-up

Appendix Table A7 reports a second prompt-sensitivity probe. For the Armington elasticity, the follow-up prompt made explicit that the target was the top-level import-versus-domestic substitution elasticity, not source-country substitution or a sector-level import-demand elasticity. Across all 17 model reruns, that clarification shifts pooled centers downward or leaves them essentially unchanged (the only increase is +0.013 for Claude Fable 5). The largest decreases appear for Gemini 3.1 Flash-Lite (-0.767), Claude Sonnet 5 (-0.647), and Claude Sonnet 4.6 and GPT-5.4 (-0.433 each). Two clarified intervals (Gemini 3 Flash, Gemini 3.1 Pro) reach exactly 0 at their lower end, which reflects the registry support floor rather than an elicited quantile. I interpret the deltas as further evidence that even apparently standard macro parameters can hide meaningful object-definition ambiguity.

Appendix Table A7. Armington clarification follow-up.

Note: Change in the Armington elasticity after clarifying that the target is the top-level import-versus-domestic elasticity, not source-country or sector-level substitution.

Model	Old center	Clarified center	Change	Old pooled 90% interval	Clarified pooled 90% interval
Claude Fable 5	1.987	2	0.013	[0.9024, 4.738]	[0.9508, 3.995]
Claude Haiku 4.5	1.5	1.387	-0.113	[0.6154, 2.979]	[0.4396, 2.787]
Claude Opus 4.7	1.533	1.5	-0.033	[0.6788, 4.574]	[0.6687, 3.983]
Claude Opus 4.8	2.4	2.033	-0.367	[0.9516, 6.965]	[0.9494, 5.661]
Claude Sonnet 4.6	2.267	1.833	-0.433	[0.8333, 6.615]	[0.7826, 4.295]
Claude Sonnet 5	3.18	2.533	-0.647	[1.161, 7.883]	[1.002, 5.995]
GPT-5.4	2.5	2.067	-0.433	[1.08, 5.997]	[0.9368, 5]
GPT-5.4 mini	2.3	2.1	-0.2	[0.8105, 8.107]	[0.8221, 6.607]
GPT-5.4 nano	2.233	1.993	-0.24	[0.836, 5.921]	[0.8427, 4.391]
GPT-5.5	2.153	1.94	-0.213	[0.7121, 6.78]	[0.6748, 6.087]
Gemini 3 Flash	1.633	1.573	-0.061	[0.6512, 5.783]	[0, 5.903]
Gemini 3.1 Flash-Lite	2.367	1.6	-0.767	[0.6797, 7.27]	[0.5283, 4.856]
Gemini 3.1 Pro	1.653	1.444	-0.209	[0.5471, 4.933]	[0, 5.592]
Gemini 3.5 Flash	1.753	1.387	-0.367	[0.6111, 5.062]	[0.5137, 3.815]
Grok 4.1 Fast	1.5	1.5	0	[0.6512, 3]	[0.7059, 3]
Grok 4.20	2	1.967	-0.033	[0.5034, 7.71]	[0.5084, 7.252]
Grok 4.3	2.94	2.7	-0.24	[0.8114, 9.02]	[0.7451, 8.039]

8.8 IES clarification follow-up

Appendix Table A8 reports a third prompt-sensitivity probe on the intertemporal elasticity of substitution. The follow-up prompt makes explicit that the target is the annual macro-calibration IES for nondurable consumption, not a generic inverse-CRRA or asset-pricing object.

Under the clarification, most pooled centers stay near 0.5 or move slightly downward; the largest shift is -0.842 for Gemini 3.1 Pro (from 1.467 to 0.625), which suggests the base prompt's IES interpretation was picking up more of the inverse-CRRA reading than the macro-calibration reading for that model. This is a narrower ambiguity than the Armington and income-sign cases, but it is conceptually important because the IES is often conflated with risk aversion in casual model discussions.

Appendix Table A8. IES clarification follow-up.

Note: Full-panel follow-up across all 11 models. Change in the intertemporal elasticity of substitution after clarifying that the target is the annual macro-calibration IES for nondurable consumption, not a generic inverse-CRRA or asset-pricing object.

Model	Old center	Clarified center	Change	Old pooled 90% interval	Clarified pooled 90% interval
Claude Fable 5	0.5	0.407	-0.093	[0.1, 1.661]	[0.0692, 1.346]
Claude Haiku 4.5	0.5	0.5	0	[0.2058, 1.188]	[0.2115, 1.193]
Claude Opus 4.7	0.5	0.5	0	[0.092, 1.965]	[0.1, 1.483]
Claude Opus 4.8	0.5	0.5	0	[0.1, 1.5]	[0.1, 1.455]
Claude Sonnet 4.6	0.5	0.5	0	[0.1, 1.5]	[0.1, 1.491]
Claude Sonnet 5	0.5	0.5	0	[0.1024, 1.482]	[0.1024, 1.199]
GPT-5.4	0.5	0.5	0	[0.1133, 1.388]	[0.1862, 1.2]
GPT-5.4 mini	0.54	0.5	-0.04	[0.1014, 1.924]	[0.092, 1.498]
GPT-5.4 nano	0.747	0.733	-0.013	[0.2167, 1.956]	[0.2118, 1.923]
GPT-5.5	0.5	0.41	-0.09	[0.06411, 1.664]	[0.04859, 1.423]
Gemini 3 Flash	0.5	0.478	-0.022	[0.1, 1.979]	[0, 1.667]
Gemini 3.1 Flash-Lite	0.5	0.5	0	[0.1, 1.431]	[0.1018, 1.198]
Gemini 3.1 Pro	1.467	0.625	-0.842	[0.2857, 2.499]	[0, 1.767]

Model	Old center	Clarified center	Change	Old pooled 90% interval	Clarified pooled 90% interval
Gemini 3.5 Flash	0.933	0.567	-0.367	[0.07895, 2.11]	[0.1037, 1.899]
Grok 4.1 Fast	0.5	0.5	0	[0.1474, 1.97]	[0.1429, 2]
Grok 4.20	0.967	0.667	-0.3	[0.2004, 3.611]	[0.1211, 3.224]
Grok 4.3	0.873	0.863	-0.01	[0.2, 2.457]	[0.2164, 2.421]

8.9 Capital-gains convention audit

Appendix Table A9 elicits both parameterizations of the capital-gains realizations elasticity under the v4 main prompt: the headline panel's w.r.t.-tax-rate convention and a sibling quantity defined w.r.t. the net-of-tax rate. The identity $\varepsilon_\tau = -\frac{\tau}{1-\tau}\,\varepsilon_{1-\tau}$ inverts to $\tau = -\rho/(1-\rho)$ where $\rho = \varepsilon_\tau/\varepsilon_{1-\tau}$. Rather than plug each model's two pooled centers into that inversion once, I bootstrap the implied-τ distribution by drawing 1,000 independent pairs $(\varepsilon_{\tau,i}, \varepsilon_{1-\tau,j})$ from each model's 15 tax-rate runs and 15 net-of-tax-rate runs and inverting per pair. The table reports the median and 90 percent interval of that distribution. The bootstrap median need not equal the plug-in ratio of medians: $\tau = -\rho/(1-\rho)$ is nonlinear in ρ, so Jensen's inequality lets averaging implied-τ values across pairs move the center relative to the implied τ at the two pooled medians, and the bootstrap quantiles also make the residual uncertainty visible.[2]

The table uses two band anchors rather than one. The narrow [0.15, 0.37] window covers the U.S. top-bracket long-term-capital-gains envelope (federal LTCG top of 0.20, plus a 0.038 NIIT, plus the highest state LTCG layer); the wider (0.37, 0.55] window covers an ordinary-income-rate anchor (federal ordinary-income top plus high state). An implied median τ inside the first window is flagged as "LTCG-rate consistent" and inside the second as "ordinary-income-rate consistent"; implied τ outside both windows but still in $(0, 1)$ is flagged as "plausible sign, outside bands"; and non-positive or greater than one is flagged as "out of band". The "shared-tau coherence" column then flags the joint sign pattern of the two pooled medians: cells outside the canonical sign-consistent pattern (tax ≤ 0, net ≥ 0) indicate that the model's two answers are not jointly consistent with any single τ — either because the model holds two independent literature anchors, or because one convention was answered with the opposite sign. Sixteen of seventeen models are sign-consistent; the exception (GPT-5.4 nano) returns

[2] A shared-tau coherence violation does not strictly demand a convention swap. A model could legitimately hold two defensible but internally independent magnitudes drawn from different literatures — for example, a net-of-tax elasticity anchored in a medium-run realizations study and a tax-rate elasticity anchored in a short-run timing study — in which case the bootstrap would still reveal joint incoherence even though neither individual answer is wrong. The audit therefore identifies joint incoherence under a single shared τ rather than labeling the underlying cause.

two negative medians, so the identity's premise fails and no implied τ is defined for it. Among the sign-consistent models, one (Gemini 3.5 Flash) has a bootstrap distribution that straddles the $\rho = 1$ pole — only 65% of its draws fall in $(0, 1)$ — and is flagged uninformative rather than assigned a band. The substantive finding concerns the remaining fifteen: eleven cluster in the ordinary-income-rate window and four in the LTCG window, so feeding the models' own responsiveness estimates into the inversion typically recovers an implied τ closer to the top marginal ordinary-income rate than to the top LTCG rate. This is the most direct internal-consistency check the design supports, since the sibling probes the exact same economic object from the other side.

Appendix Table A9. Capital-gains convention audit.

Note: Both capital-gains-realizations conventions elicited independently under prompt v4. Under the identity epsilon_taxrate = -(tau / (1 - tau)) * epsilon_netoftax, any model whose two answers are jointly consistent with a single tau prior implies one specific tau. The implied-tau column reports the median and 90 percent interval of 1000 bootstrap draws that independently sample one tax-rate run and one net-of-tax-rate run from the 15 runs of each; this is not the plug-in ratio of medians, which differs from the bootstrap median because tau = -rho / (1 - rho) is nonlinear in rho (Jensen's inequality). The band flag is computed against the bootstrap median. 'LTCG-rate consistent' marks an implied median tau in [0.15, 0.37] (U.S. top-bracket long-term-capital-gains envelope, federal LTCG plus NIIT plus high state); 'ordinary-income-rate consistent' marks (0.37, 0.55] (federal ordinary-income top plus high state); 'plausible sign, outside bands' marks (0, 1) outside both windows; and 'out of band' marks a non-positive or > 1 median tau. The shared-tau coherence column flags the joint sign pattern of the two pooled medians: a model with (tax < 0, net > 0) is sign-consistent with the identity, whereas (both positive, both negative, or reversed) indicates that the two answers are not jointly consistent with a single tau prior — either because the model holds two independent literature anchors, or because one convention was answered with the opposite sign.

Model	Provider	epsilon w.r.t. tax rate (median)	epsilon w.r.t. net-of-tax rate (median)	Implied tau median [90%]	Share of draws in (0, 1)	Shared-tau coherence	Band (LTCG [0.15, 0.37], ordinary-income [0.37, 0.55])
GPT-5.4 nano	OpenAI	-0.35	-0.2	— (premise fails)	—	both negative	not identified

Model	Provider	epsilon w.r.t. tax rate (median)	epsilon w.r.t. net-of-tax rate (median)	Implied tau median [90%]	Share of draws in (0, 1)	Shared-tau coherence	Band (LTCG [0.15, 0.37], ordinary-income [0.37, 0.55])
Claude Fable 5	Anthropic	-0.7	1.8	0.280 [0.250, 0.318]	100%	sign-consistent (tax<0, net>0)	LTCG-rate consistent
Claude Haiku 4.5	Anthropic	-0.8	0.8	0.500 [0.348, 0.552]	100%	sign-consistent (tax<0, net>0)	ordinary-income-rate consistent
Claude Opus 4.7	Anthropic	-0.7	0.7	0.500 [0.500, 0.500]	100%	sign-consistent (tax<0, net>0)	ordinary-income-rate consistent
Claude Opus 4.8	Anthropic	-0.7	0.7	0.500 [0.500, 0.538]	100%	sign-consistent (tax<0, net>0)	ordinary-income-rate consistent
Claude Sonnet 4.6	Anthropic	-0.7	3.5	0.167 [0.167, 0.189]	100%	sign-consistent (tax<0, net>0)	LTCG-rate consistent
Claude Sonnet 5	Anthropic	-0.6	0.6	0.500 [0.500, 0.538]	100%	sign-consistent (tax<0, net>0)	ordinary-income-rate consistent
GPT-5.4	OpenAI	-0.7	0.9	0.450 [0.389, 0.571]	100%	sign-consistent (tax<0, net>0)	ordinary-income-rate consistent

Model	Provider	epsilon w.r.t. tax rate (median)	epsilon w.r.t. net-of-tax rate (median)	Implied tau median [90%]	Share of draws in (0, 1)	Shared-tau coherence	Band (LTCG [0.15, 0.37], ordinary-income [0.37, 0.55])
GPT-5.4 mini	OpenAI	-0.8	1.2	0.500 [0.333, 0.636]	100%	sign-consistent (tax<0, net>0)	ordinary-income-rate consistent
GPT-5.5	OpenAI	-0.65	0.9	0.400 [0.207, 0.519]	100%	sign-consistent (tax<0, net>0)	ordinary-income-rate consistent
Gemini 3 Flash	Google	-0.8	0.8	0.500 [0.467, 0.515]	100%	sign-consistent (tax<0, net>0)	ordinary-income-rate consistent
Gemini 3.1 Flash-Lite	Google	-0.75	0.8	0.484 [0.318, 0.538]	100%	sign-consistent (tax<0, net>0)	ordinary-income-rate consistent
Gemini 3.1 Pro	Google	-0.7	2	0.259 [0.200, 0.483]	100%	sign-consistent (tax<0, net>0)	LTCG-rate consistent
Gemini 3.5 Flash	Google	-0.4	0.8	0.452 [-0.889, 3.250]	65%	sign-consistent (tax<0, net>0)	uninformative (pole-straddling)
Grok 4.1 Fast	xAI	-0.7	1.2	0.368 [0.149, 0.500]	100%	sign-consistent (tax<0, net>0)	LTCG-rate consistent

Model	Provider	epsilon w.r.t. tax rate (median)	epsilon w.r.t. net-of-tax rate (median)	Implied tau median [90%]	Share of draws in (0, 1)	Shared-tau coherence	Band (LTCG [0.15, 0.37], ordinary-income [0.37, 0.55])
Grok 4.20	xAI	-0.5	0.65	0.417 [0.348, 0.478]	100%	sign-consistent (tax<0, net>0)	ordinary-income-rate consistent
Grok 4.3	xAI	-0.7	0.8	0.484 [0.411, 0.556]	100%	sign-consistent (tax<0, net>0)	ordinary-income-rate consistent

9 Appendix: Simulation-Facing Coefficients

The simulation-facing substitution-response coefficients are substantively useful, but they are implementation-facing response parameters rather than textbook elasticity objects. I therefore keep them out of the headline rankings and report them separately here.

Appendix Table A10. Simulation-facing model overview.

Note: Simulation-facing subpanel only: 12 PolicyEngine-style substitution-response coefficients used in a U.S. tax-benefit microsimulation. These rows are reported separately from the canonical elasticity panel because they are implementation-facing response parameters rather than standard elasticity objects.

Model	Provider	Avg abs-elasticity rank (1=highest)	Avg predictive-uncertainty rank (1=narrowest)	Mean absolute pooled center	Mean pooled 90% width	Success rate	Cost / successful run
GPT-5.4 nano	OpenAI	1.12	16.67	0.507	1.319	100.0%	$0.0003

Model	Provider	Avg abs-elasticity rank (1=highest)	Avg predictive-uncertainty rank (1=narrowest)	Mean absolute pooled center	Mean pooled 90% width	Success rate	Cost / successful run
Grok 4.20	xAI	4.08	13	0.277	0.765	100.0%	$0.0086
Grok 4.1 Fast	xAI	4.54	16	0.295	1.219	100.0%	$0.0002
Claude Opus 4.7	Anthropic	5.33	10.25	0.267	0.637	100.0%	$0.0216
Claude Opus 4.8	Anthropic	5.54	10.42	0.267	0.626	100.0%	$0.0141
Grok 4.3	xAI	6.42	12	0.262	0.681	100.0%	$0.0017
Claude Haiku 4.5	Anthropic	7.83	12.17	0.247	0.678	100.0%	$0.0031
Claude Sonnet 4.6	Anthropic	8	7.33	0.241	0.522	100.0%	$0.0109
Claude Sonnet 5	Anthropic	9.21	10	0.228	0.631	100.0%	$0.0073
Claude Fable 5	Anthropic	9.5	2.58	0.222	0.42	100.0%	$0.0468
GPT-5.4 mini	OpenAI	10.04	13.58	0.218	0.78	100.0%	$0.0010
Gemini 3.1 Pro	Google	10.25	3	0.216	0.432	100.0%	$0.0078
GPT-5.4	OpenAI	12.58	5.08	0.18	0.484	100.0%	$0.0036
GPT-5.5	OpenAI	13.79	7.92	0.17	0.54	100.0%	$0.0137
Gemini 3 Flash	Google	14.46	5.83	0.163	0.503	100.0%	$0.0010
Gemini 3.5 Flash	Google	14.54	3.08	0.165	0.446	100.0%	$0.0100
Gemini 3.1 Flash-Lite	Google	15.75	4.08	0.13	0.456	100.0%	$0.0008

Appendix Table A11. Simulation-facing disagreement.

Note: Simulation-facing PolicyEngine substitution-response coefficients only, sorted by cross-model spread in pooled point estimates.

Quantity	Lowest model	Lowest center	Highest model	Highest center	Spread	Mean pooled 90% width	Spread / mean width
Primary-earner substitution elasticity in a tax-benefit simulation, decile 7	Gemini 3.1 Flash-Lite	0.086	GPT-5.4 nano	0.623	0.537	0.601	0.894
Primary-earner substitution elasticity in a tax-benefit simulation, decile 8	Gemini 3.1 Flash-Lite	0.11	GPT-5.4 nano	0.609	0.499	0.619	0.805
Primary-earner substitution elasticity in a tax-benefit simulation, decile 6	Gemini 3.1 Flash-Lite	0.116	GPT-5.4 nano	0.59	0.474	0.6	0.79

Quantity	Lowest model	Lowest center	Highest model	Highest center	Spread	Mean pooled 90% width	Spread / mean width
Primary-earner substitution elasticity in a tax-benefit simulation, decile 5	Gemini 3 Flash	0.121	GPT-5.4 nano	0.593	0.473	0.608	0.778
Primary-earner substitution elasticity in a tax-benefit simulation, decile 10	Gemini 3.5 Flash	0.14	GPT-5.4 nano	0.603	0.463	0.667	0.694
Primary-earner substitution elasticity in a tax-benefit simulation, decile 9	Gemini 3.1 Flash-Lite	0.129	GPT-5.4 nano	0.587	0.458	0.664	0.689

Quantity	Lowest model	Lowest center	Highest model	Highest center	Spread	Mean pooled 90% width	Spread / mean width
Primary-earner substitution elasticity in a tax-benefit simulation, decile 3	Gemini 3.1 Flash-Lite	0.069	GPT-5.4 nano	0.52	0.451	0.608	0.742
Primary-earner substitution elasticity in a tax-benefit simulation, decile 4	Gemini 3.1 Flash-Lite	0.073	GPT-5.4 nano	0.465	0.391	0.599	0.654
Primary-earner substitution elasticity in a tax-benefit simulation, decile 1	Gemini 3.1 Flash-Lite	0.05	GPT-5.4 nano	0.343	0.293	0.62	0.473

Quantity	Lowest model	Lowest center	Highest model	Highest center	Spread	Mean pooled 90% width	Spread / mean width
Primary-earner substitution elasticity in a tax-benefit simulation, decile 2	Gemini 3.1 Flash-Lite	0.084	GPT-5.4 nano	0.376	0.292	0.621	0.47
Secondary-earner substitution elasticity in a tax-benefit simulation	Gemini 3.5 Flash	0.29	Claude Sonnet 4.6	0.5	0.21	0.943	0.223
Substitution elasticity of labor supply in a tax-benefit simulation	Gemini 3.1 Flash-Lite	0.153	GPT-5.4 nano	0.307	0.153	0.714	0.215

9.1 Static flat-tax plus demogrant frontier

Appendix Table A12 reports a static blank-slate flat-tax benchmark on the same Enhanced CPS 2024 microdata (PolicyEngine Team 2024). Each row applies a flat tax to PolicyEngine's current positive-AGI base and rebates the resulting revenue as an equal per-person demogrant using tax-unit size. This is the modernized analogue of a blank-slate UBI-style exercise in the current PolicyEngine stack.

I do not present it as an optimal-tax result. The benchmark holds behavior fixed, omits leisure, and therefore turns into a pure distributional comparison. Mean per-person resources are mechanically constant across rows, so the informative objects are the demogrant, the distributional quantiles, and the Gini of per-person post-tax positive-AGI resources.

Appendix Table A12. Static flat positive-AGI tax plus demogrant frontier.

Note: Static PolicyEngine benchmark on Enhanced CPS 2024 microdata. Each row applies a flat tax to the current positive-AGI base and rebates the revenue as an equal per-person demogrant using tax-unit size. This is a distributional frontier, not a behavioral or leisure-adjusted optimal-tax exercise. Mean per-person resources are mechanically constant across rows, so the informative objects are the demogrant, the distributional quantiles, and the Gini of per-person post-tax positive-AGI resources.

Flat tax rate	Demogrant per person	P10 post-tax resources	Median post-tax resources	P90 post-tax resources	Gini
0%	$0	$1,554	$24,091	$85,300	0.627
20%	$8,881	$10,125	$28,154	$77,121	0.501
40%	$17,763	$18,695	$32,217	$68,942	0.376
60%	$26,644	$27,266	$36,280	$60,764	0.251
80%	$35,525	$35,836	$40,343	$52,585	0.125
95%	$42,186	$42,264	$43,391	$46,451	0.031

9.2 Top-rate robustness to Pareto tail and CRRA curvature

Appendix Table A13 reports the top-rate mapping from Table 4 under alternative values of the Pareto tail parameter a and the CRRA curvature γ. The column-wise comparison asks how the implied optimal top rate moves if we replace the microdata-calibrated $a = 1.621$ with a Pareto tail at 1.3, 1.5, or 1.7, or if we replace log utility with CRRA at $\gamma = 2$ while holding a at the microdata estimate. The ETI median feeding each row is the same pooled-mixture median used in Table 4, so column-wise differences reflect only the formula's (a, γ) pair and not any change in the elicited responses. Across all parameterizations, the cross-model ordering is unchanged.

Appendix Table A13. Top-rate robustness to Pareto tail and CRRA curvature.

Note: Robustness of the utilitarian optimal top-rate mapping in Table 4 to the Pareto tail parameter a and the CRRA coefficient gamma. Each cell is the median implied optimal top rate tau* = (1 - g_bar) / (1 - g_bar + a e) computed at the model's pooled ETI median under the (a, gamma) pair in the column header, where g_bar = a / (a + gamma). The baseline column (a = 1.621, gamma = 1) reproduces the median column in Table 4. The a = 1.3 / 1.5 / 1.7 columns vary only the Pareto tail while keeping log utility (gamma = 1); the final column replaces log utility with CRRA at gamma = 2 while holding a at the baseline value above. Under log utility the corresponding welfare weights g_bar are 1.3/2.3 = 0.565, 1.621/(1 + 1.621) = 0.618, 1.5/2.5 = 0.600, and 1.7/2.7 = 0.630; under gamma = 2 with a = 1.621, g_bar = 1.621/(1.621 + 2) = 0.448.

Model	ETI median	Baseline top rate (a=1.621, gamma=1)	Top rate (a=1.3, gamma=1)	Top rate (a=1.5, gamma=1)	Top rate (a=1.7, gamma=1)	Top rate (a=1.621, gamma=2)
Gemini 3.1 Pro	0.351	40.2%	48.8%	43.2%	38.3%	49.3%
Gemini 3.5 Flash	0.357	39.8%	48.4%	42.8%	37.9%	48.9%
GPT-5.5	0.369	39.0%	47.6%	42.0%	37.1%	48.0%
Claude Opus 4.8	0.383	38.0%	46.6%	41.0%	36.2%	47.1%
Gemini 3.1 Flash-Lite	0.389	37.7%	46.2%	40.7%	35.9%	46.7%
Claude Opus 4.7	0.4	37.0%	45.5%	40.0%	35.3%	46.0%
Gemini 3 Flash	0.4	37.0%	45.5%	40.0%	35.3%	46.0%
Grok 4.1 Fast	0.4	37.0%	45.5%	40.0%	35.3%	46.0%
GPT-5.4	0.42	35.9%	44.3%	38.8%	34.2%	44.8%
Claude Fable 5	0.437	35.0%	43.4%	37.9%	33.3%	43.8%
GPT-5.4 mini	0.437	35.0%	43.4%	37.9%	33.3%	43.8%
Grok 4.3	0.439	34.9%	43.3%	37.8%	33.2%	43.7%
Claude Sonnet 5	0.471	33.3%	41.5%	36.1%	31.6%	42.0%
Claude Sonnet 4.6	0.5	32.0%	40.1%	34.8%	30.3%	40.5%
Grok 4.20	0.5	32.0%	40.1%	34.8%	30.3%	40.5%
Claude Haiku 4.5	0.502	31.9%	40.0%	34.7%	30.3%	40.4%
GPT-5.4 nano	0.546	30.1%	38.0%	32.8%	28.5%	38.4%

9.3 Resampling standard errors and rank stability

Appendix Table A14 resamples the 15 runs within each canonical cell (200 bootstrap resamples, fixed seed) to attach Monte Carlo standard errors to the pooled center and pooled 90 percent width, and to test whether the predictive-uncertainty ordering survives run-level resampling. Median center standard errors are small in absolute terms for every model, relative width standard errors run roughly 2 to 5 percent, and each model's average width rank carries a narrow 90 percent resampling interval. The width ordering reported in the main text is therefore signal rather than $R = 15$ noise. This diagnostic covers the canonical-panel width ordering; the subpanel top-three and bottom-three callouts in Tables 1-2 average over fewer quantities and carry correspondingly wider resampling bands, so they should be read as coarse groupings rather than exact placements.

Appendix Table A14. Monte Carlo resampling of pooled summaries.

Note: Monte Carlo uncertainty in the pooled summaries from resampling the 15 runs within each canonical cell (200 bootstrap resamples, fixed seed). Center MC SE is the standard error of the pooled center; relative width MC SE is the standard error of the pooled 90% width divided by its mean. The final columns show the distribution of each model's average width rank across resamples; narrow intervals indicate the predictive-uncertainty ordering is stable to run-level resampling at R = 15.

Model	Median center MC SE	Median relative width MC SE	Avg width rank (mean)	Avg width rank 90% interval
Claude Fable 5	0.003	3%	5.32	[4.84, 5.77]
Claude Haiku 4.5	0.005	5%	6.2	[5.54, 6.85]
Claude Sonnet 5	0.002	2%	6.55	[6.15, 6.92]
Claude Opus 4.7	0	2%	6.69	[6.23, 7.15]
Gemini 3.1 Pro	0.006	2%	7.12	[6.62, 7.54]
Claude Opus 4.8	0	1%	7.46	[7.00, 7.92]
Gemini 3 Flash	0	1%	7.74	[7.30, 8.15]
Gemini 3.5 Flash	0.01	3%	8.34	[7.92, 8.69]
Gemini 3.1 Flash-Lite	0.006	5%	8.65	[8.00, 9.31]
Claude Sonnet 4.6	0	1%	9.11	[8.69, 9.62]
GPT-5.5	0.005	2%	9.5	[9.15, 9.77]
GPT-5.4	0	3%	9.61	[9.08, 10.00]
GPT-5.4 nano	0.023	6%	10.94	[10.61, 11.31]
Grok 4.1 Fast	0	2%	11.04	[10.15, 11.69]
Grok 4.3	0.012	5%	11.44	[10.85, 11.93]
GPT-5.4 mini	0.009	5%	13.41	[13.00, 13.85]

Model	Median center MC SE	Median relative width MC SE	Avg width rank (mean)	Avg width rank 90% interval
Grok 4.20	0.011	4%	13.88	[13.38, 14.31]

9.4 Within-run versus between-run variance

Appendix Table A15 splits pooled predictive variance into its two components: the within-run term (the model's own stated p05-p95 quantiles) and the between-run term (variation of run means across repeated draws). The between-run share is a median of 0 to 2 percent for every model, so the pooled widths — and the predictive-uncertainty ranking built on them — are dominated by what the models state, not by sampling noise. The maximum single-cell shares are larger — 29% for gpt-5.4-nano and 25% for gemini-3.5-flash (no other model exceeds 7%), concentrated in the sign-unstable cells discussed above — so the median, not the maximum, characterizes the typical cell. This also bounds the concern that the three no-sampling-parameter Claude models are ranked under a different draw regime: the component their regime affects is a negligible share of the total for every model in the panel. (The between-run term uses the population variance over the 15 run means, a mild downward bias that is immaterial given these shares; the headline interval is the nonparametric mixture, not a Gaussian built from the summed variance.)

Appendix Table A15. Variance decomposition of pooled predictive spread.

Note: Split of the pooled predictive variance over the canonical 13-quantity subpanel into the within-run component (the model's own stated p05-p95 quantiles) and the between-run component (variation of run means across the 15 repeated draws). The between-run share is the only component affected by the sampling regime, which differs for the three no-sampling-parameter Claude models.

Model	Cells	Median within-run SD	Median between-run SD	Median between-run variance share	Max between-run variance share
Claude Fable 5	13	0.679	0.012	0%	1%
Claude Haiku 4.5	13	0.672	0.018	0%	4%
Claude Opus 4.7	13	0.704	0	0%	0%
Claude Opus 4.8	13	0.72	0	0%	1%
Claude Sonnet 4.6	13	0.726	0	0%	1%

Model	Cells	Median within-run SD	Median between-run SD	Median between-run variance share	Max between-run variance share
Claude Sonnet 5	13	0.699	0.01	0%	1%
GPT-5.4	13	0.684	0	0%	1%
GPT-5.4 mini	13	0.713	0.034	1%	5%
GPT-5.4 nano	13	0.699	0.098	2%	29%
GPT-5.5	13	0.703	0.018	0%	1%
Gemini 3 Flash	13	0.677	0	0%	1%
Gemini 3.1 Flash-Lite	13	0.687	0.025	0%	3%
Gemini 3.1 Pro	13	0.689	0.025	0%	3%
Gemini 3.5 Flash	13	0.683	0.041	0%	25%
Grok 4.1 Fast	13	0.743	0	0%	4%
Grok 4.20	13	0.706	0.042	0%	7%
Grok 4.3	13	0.68	0.046	0%	6%

9.5 Harness disclosure

Appendix Table A16 reports the full generation-harness configuration per model. The prompt text and repeated-run design are identical across models; the structured-output mechanism, completion budget, sampling regime, and reasoning configuration follow each provider's API surface and are therefore confounded with model identity. Completion budgets are truncation guards rather than elicitation content — reasoning tokens count against them on models that reason, so budgets were raised where required to avoid truncation. One further asymmetry: the OpenAI path wraps the elicitation prompt with a one-line system message ("Follow the user's instructions exactly and return only the final answer."), while the Anthropic and LiteLLM paths send the prompt as a bare user message; the instruction is format-only and is attached unconditionally by the OpenAI request builder (the committed request logs record usage metadata rather than message payloads, so the string is checkable in llm_econ_beliefs/providers.py).

Appendix Table A16. Per-model generation-harness configuration.

Note: Per-model generation-harness configuration. The prompt text and repeated-run design are identical across models; the structured-output mechanism, completion budget, sampling regime, and reasoning configuration follow each provider's API surface and are therefore confounded with model identity. Completion budgets are truncation guards: reasoning tokens count against them on models that reason, so budgets were raised where required to avoid truncation. Identifiers marked alias float with provider updates; dated snapshots are pinned.

Model	Provider path	Output mechanism	Completion budget	Sampling	Reasoning config	API identifier	Identifier type
GPT-5.5	OpenAI Chat Completions	strict JSON schema	1200 (8000 for the 40 re-elicited runs)	temperature 1.0, batched n <= 8	provider default effort	gpt-5.5	alias
GPT-5.4	OpenAI Chat Completions	strict JSON schema	1200	temperature 1.0, batched n <= 8	provider default effort	gpt-5.4	alias
GPT-5.4 mini	OpenAI Chat Completions	strict JSON schema	1200	temperature 1.0, batched n <= 8	provider default effort	gpt-5.4-mini	alias
GPT-5.4 nano	OpenAI Chat Completions	strict JSON schema	1200	temperature 1.0, batched n <= 8	provider default effort	gpt-5.4-nano	alias
Claude Fable 5	native Anthropic API	strict JSON schema	32000	none accepted (provider default)	always-on reasoning	claude-fable-5	alias
Claude Opus 4.8	native Anthropic API	strict JSON schema	32000	none accepted (provider default)	off (provider default)	claude-opus-4-8	alias
Claude Sonnet 5	native Anthropic API	strict JSON schema	32000	none accepted (provider default)	adaptive (provider default)	claude-sonnet-5	alias
Claude Opus 4.7	LiteLLM	forced function call	1200	temperature 1.0	off (provider default)	claude-opus-4-7	alias

Model	Provider path	Output mechanism	Completion budget	Sampling	Reasoning config	API identifier	Identifier type
Claude Sonnet 4.6	LiteLLM	forced function call	1200	temperature 1.0	off (provider default)	claude-sonnet-4-6	alias
Claude Haiku 4.5	LiteLLM	forced function call	1200	temperature 1.0	off (provider default)	claude-haiku-4-5-20251001	dated snapshot
Gemini 3.1 Pro	LiteLLM	forced JSON object	1200	temperature 1.0	provider default thinking	gemini-3.1-pro-preview	preview alias
Gemini 3.5 Flash	LiteLLM	forced JSON object	4000	temperature 1.0	provider default thinking	gemini-3.5-flash	alias
Gemini 3 Flash	LiteLLM	forced JSON object	1200	temperature 1.0	provider default thinking	gemini-3-flash-preview	preview alias
Gemini 3.1 Flash-Lite	LiteLLM	forced JSON object	1200	temperature 1.0	provider default thinking	gemini-3.1-flash-lite-preview	preview alias
Grok 4.20	LiteLLM	forced function call	1200	temperature 1.0	reasoning variant	xai/grok-4.20-reasoning	alias
Grok 4.3	LiteLLM	forced function call	4000	temperature 1.0	provider default	xai/grok-4.3	alias
Grok 4.1 Fast	LiteLLM	forced function call	1200	temperature 1.0	non-reasoning variant	xai/grok-4-1-fast-non-reasoning	alias

9.6 Cross-mechanism ablation

Appendix Table A17 addresses the mechanism confound directly by re-eliciting Claude Opus 4.7 — an April model originally run through the LiteLLM forced-function-call path at a 1,200-token budget — through the July native strict-JSON-schema path at a 32,000-token budget, on the nine canonical elasticities with 15 fresh runs each. Pooled centers move by at most 0.03 (median 0.00); pooled widths move more, from -15 to +7 percent across the nine quantities with no consistent direction. Within the panel's resolution, the harness mechanism does not move elicited centers, which supports reading cross-wave differences in centers as model differences; width comparisons across waves carry the extra mechanism noise. The ablation isolates the output mechanism and completion budget only — Claude Opus 4.7 runs without extended reasoning on both paths, so the July models' always-on or adaptive reasoning modes remain confounded with model identity.

Appendix Table A17. Same model, two harness mechanisms.

Note: Same model (Claude Opus 4.7), same v4 prompts, same repeated-run design, two harness mechanisms: the April LiteLLM forced-function-call path (temperature 1.0, 1200-token budget) versus the July native Anthropic strict-JSON-schema path (no sampling parameters, 32000-token budget). Each cell pools 15 fresh runs elicited in July 2026 under the native mechanism against the April panel cell. Centers are stable (max absolute change 0.03); widths move from -15 to +7 percent across quantities with no consistent direction. Opus 4.7 runs without extended reasoning on both paths, so the reasoning-mode axis is not covered.

Quantity	LiteLLM center	Native center	Center change	LiteLLM 90% width	Native 90% width
Armington elasticity	1.533	1.533	0	3.895	4.057
Capital gains realizations elasticity	-0.7	-0.7	0	1.527	1.63
Elasticity of substitution between capital and labor	0.6	0.6	0	0.903	0.885
Elasticity of taxable income	0.4	0.4	0	0.873	0.892
Employment participation elasticity of single mothers	0.7	0.71	0.01	1.196	1.198
Frisch elasticity of labor supply	0.417	0.387	-0.03	1.318	1.119
Income elasticity of labor supply	-0.05	-0.05	0	0.22	0.22

Quantity	LiteLLM center	Native center	Center change	LiteLLM 90% width	Native 90% width
Intertemporal elasticity of substitution	0.5	0.5	0	1.873	1.698
Uncompensated wage elasticity of labor supply	0.1	0.1	0	0.449	0.4

9.7 Support bounds registry

Appendix Table A18 tabulates the per-quantity support bounds that drive quantile-bin reconstruction, tail extrapolation, and the transforms behind the REML and Bayesian estimators. For unbounded sides, the reconstruction extends the support 0.25 x the adjacent inter-quantile gap beyond the elicited p05/p95; the headline pooled 5-95 interval is largely insensitive to this rule because its endpoints sit at or near the elicited quantile knots, while the SD-based and REML/Bayes summaries are more exposed to it.

Appendix Table A18. Registry support bounds.

Note: Registry support bounds used for quantile-bin reconstruction, tail extrapolation, and the transforms behind the REML and Bayesian estimators. Unbounded sides use a tail-extrapolation rule that extends 0.25 x the adjacent inter-quantile gap beyond the elicited p05/p95.

Quantity	Quantity id	Lower support	Upper support
Annual discount factor	household.annual_discount_factor	0	1
Intertemporal elasticity of substitution	household.intertemporal_elasticity_of_substitution	0	5
Coefficient of relative risk aversion	household.relative_risk_aversion.crra	0	50
Employment participation elasticity of single mothers	labor_supply.extensive_margin.single_mothers	-1	5

Quantity	Quantity id	Lower support	Upper support
Frisch elasticity of labor supply	labor_supply.frisch_elasticity.prime_age	0	10
Income elasticity of labor supply	labor_supply.income_elasticity.prime_age	-2	1
Uncompensated wage elasticity of labor supply	labor_supply.marshallian_wage_elasticity.prime_age	-2	4
Substitution elasticity of labor supply in a tax-benefit simulation	labor_supply.policy_response.substitution_elasticity.all	0	5
Primary-earner substitution elasticity in a tax-benefit simulation, decile 1	labor_supply.policy_response.substitution_elasticity.primary.decile_1	0	5
Primary-earner substitution elasticity in a tax-benefit simulation, decile 10	labor_supply.policy_response.substitution_elasticity.primary.decile_10	0	5
Primary-earner substitution elasticity in a tax-benefit simulation, decile 2	labor_supply.policy_response.substitution_elasticity.primary.decile_2	0	5
Primary-earner substitution elasticity in a tax-benefit simulation, decile 3	labor_supply.policy_response.substitution_elasticity.primary.decile_3	0	5
Primary-earner substitution elasticity in a tax-benefit simulation, decile 4	labor_supply.policy_response.substitution_elasticity.primary.decile_4	0	5
Primary-earner substitution elasticity in a tax-benefit simulation, decile 5	labor_supply.policy_response.substitution_elasticity.primary.decile_5	0	5
Primary-earner substitution elasticity in a tax-benefit simulation, decile 6	labor_supply.policy_response.substitution_elasticity.primary.decile_6	0	5

Quantity	Quantity id	Lower support	Upper support
Primary-earner substitution elasticity in a tax-benefit simulation, decile 7	labor_supply.policy_response.substitution_elasticity.primary.decile_7	0	5
Primary-earner substitution elasticity in a tax-benefit simulation, decile 8	labor_supply.policy_response.substitution_elasticity.primary.decile_8	0	5
Primary-earner substitution elasticity in a tax-benefit simulation, decile 9	labor_supply.policy_response.substitution_elasticity.primary.decile_9	0	5
Secondary-earner substitution elasticity in a tax-benefit simulation	labor_supply.policy_response.substitution_elasticity.secondary	0	5
TFP persistence	macro.tfp_persistence.ar1	0	1
Elasticity of substitution between capital and labor	production.capital_labor_substitution	0	10
Capital share in production	production.capital_share	0	1
Capital gains realizations elasticity	tax.capital_gains_realizations.elasticity	-10	2
Capital gains realizations elasticity (net-of-tax-rate convention)	tax.capital_gains_realizations.elasticity.net_of_tax_rate	-2	10
Elasticity of taxable income	tax.elasticity_of_taxable_income.top_earners	-1	5
Armington elasticity	trade.armington_elasticity.import_domestic	0	20

10 Statements

Data accessibility. All elicitation code, raw run-level responses, request logs, generated tables, and the scripts that rebuild every table in this paper are public at https://github.com/PolicyEngine/llm-econ-beliefs. The cached-results reproduction path rebuilds all paper tables without provider API access; full re-elicitation instructions and tracked costs are in the repository README. A versioned archive with a DOI will be deposited on acceptance.

Author contributions. M.G. designed the study, collected the data, performed the analysis, and wrote the paper.

Competing interests. The author is a co-founder of PolicyEngine, whose tax-benefit microsimulation parameters are among the simulation-facing quantities studied in this paper.

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Ethics. This study involved no human participants or animal subjects; the elicited subjects are commercial language-model APIs.

Use of AI. Language models are the object of study throughout. AI coding assistants were used in developing the elicitation harness and drafting portions of the manuscript; all analyses and claims were verified by the author.

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